KAN Komite Akreditasi Nasional
LK-002-IDN

TELKOM TEST HOUSE
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Report No.: 103/KAL/2025

External Calibration Certificate

Applicant's Data	:	Laboratorium Telkom Test House (TTH) Jl. Gegerkalong Hilir No. 47 Bandung 40152.

Equipment Data

Equipment name	:	OPTICAL ATTENUATOR
Brand	:	EXFO
Type/Model	:	3500
Serial number	:	1638775
Asset Number	:	216/CAL

Calibration Information

Delivery Date	:	May 28, 2025
Calibration By	:	Lab Kalibrasi Telkom Jakarta (LK-016-IDN)
Certificate Number	:	JCL/25/0350/HS
Calibration Date	:	May 28, 2025
Calibration Due	:	May 28, 2026

Bandung, 18 Juli 2025

Agreed by:	Reviewed by:
I GEDE ASTAWA	**AHMAD ARIF RAHMAN**
SM Infrastructure Service Research & Assurance	Mgr. Standardization & Suitability Assurance

No. Dokumen : Form TCL-QP-19-01
No. / Tgl. Terbit : 01 / 20 Juli 2017
No. / Tgl. Revisi : 02 / 04 Januari 2021

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CERTIFICATE OF CALIBRATION

Certificate Number
JCL/25/0350/HS

P T   T E L K O M   I N D O N E S I A   T b k

KAN ACCREDITED CALIBRATION LABORATORY (LK-016-IDN)

Approved Signatory

Jakarta : Jl. Percetakan Negara No.17-19 Jakarta Pusat 10570 Phone 021-21480341
Medan : Jl. Gaharu No. 1 Medan 20235 Phone 061-4531311
Surabaya : Jl. Gayungan PTT No. 17-19 Surabaya 60235 Phone 031-8283678
Makassar : Jl. AP Pettarani No. 4 Makassar 90221 Phone 0411-861153

CS Support
081-16-4-17025

M. Ilyas

Date of Issue	: 02-Jun-2025
Submitted by	: PT. TELKOM INDONESIA, Tbk. Laboraturium Telkom Test House (TTH)
Attention	: Mgr. Standardization & Suitability Assurance
Description	: Variable Optical Attenuator
Manufacturer/Model	: FTBx-3500-BI-EA
Serial Number	: 1638775 (216/CAL)
Environment	: 23 °C ± 0.1 °C 57 % ± 0.5 % RH
Calibration Location	: Onsite
Date Calibrated	: 28-May-2025
Date Received	: 27-May-2025

PT. Telkom Indonesia, Tbk. - Telkom Certifies that this instrument has been calibrated under the environmental condition stated above using telkom standard (s) which is (are) traceable to the international system of Units (SI) through SCL Hong Kong.

Method of Calibration :

The calibration was carried out based on the established Standard Telkom Calibration Procedure, TCL-WI-0013 and by comparison against the following standard(s) and instrument(s) which is (are) calibrated on schedule to maintain traceability and required accuracy :

	Description	S/N	Cal Date	Due Date
1.	Optical Fiber Power Meter	1030788-CH1	11-Apr-2025	10-Apr-2028
2.	Variable Optical Broadband Source	1082690	23-May-2025	22-May-2026

1. **LINEARITY TEST**

1.1. **LINEARITY RESULT (1310.0 nm)**

STEP ATTENUATION 3 dB

No.	UUT SETTING (dB)	MEASURED VALUE	CORRECTION	UNCERTAINTY
1	1.110	1.292 dB	0.182 dB	± 0.25 dB
2	4.110	4.328 dB	0.218 dB	± 0.25 dB
3	7.110	7.341 dB	0.231 dB	± 0.25 dB
4	10.110	10.337 dB	0.227 dB	± 0.25 dB
5	13.110	13.329 dB	0.219 dB	± 0.25 dB
6	16.110	16.328 dB	0.218 dB	± 0.25 dB
7	19.110	19.316 dB	0.206 dB	± 0.25 dB
8	22.110	22.312 dB	0.202 dB	± 0.25 dB
9	25.110	25.315 dB	0.205 dB	± 0.25 dB
10	28.110	28.313 dB	0.203 dB	± 0.25 dB
11	31.110	31.297 dB	0.187 dB	± 0.25 dB
12	34.110	34.290 dB	0.180 dB	± 0.25 dB
13	37.110	37.278 dB	0.168 dB	± 0.25 dB
14	40.110	40.282 dB	0.172 dB	± 0.25 dB
15	43.110	43.281 dB	0.171 dB	± 0.25 dB
16	46.110	46.289 dB	0.179 dB	± 0.25 dB
17	49.110	49.286 dB	0.176 dB	± 0.25 dB
18	52.110	52.279 dB	0.169 dB	± 0.25 dB

1.2. **LINEARITY RESULT (1550.0 nm)**

STEP ATTENUATION 3 dB

No.	UUT SETTING (dB)	MEASURED VALUE	CORRECTION	UNCERTAINTY
1	0.967	1.393 dB	0.426 dB	± 0.25 dB
2	3.967	4.427 dB	0.460 dB	± 0.25 dB
3	6.967	7.435 dB	0.468 dB	± 0.25 dB
4	9.967	10.438 dB	0.471 dB	± 0.25 dB
5	12.967	13.431 dB	0.464 dB	± 0.25 dB
6	15.967	16.426 dB	0.459 dB	± 0.25 dB
7	18.967	19.412 dB	0.445 dB	± 0.25 dB
8	21.967	22.414 dB	0.447 dB	± 0.25 dB
9	24.967	25.410 dB	0.443 dB	± 0.25 dB
10	27.967	28.414 dB	0.447 dB	± 0.25 dB
11	30.967	31.403 dB	0.436 dB	± 0.25 dB
12	33.967	34.409 dB	0.442 dB	± 0.25 dB
13	36.967	37.411 dB	0.444 dB	± 0.25 dB
14	39.967	40.419 dB	0.452 dB	± 0.25 dB
15	42.967	43.425 dB	0.458 dB	± 0.25 dB
16	45.967	46.427 dB	0.460 dB	± 0.25 dB
17	48.967	49.441 dB	0.474 dB	± 0.25 dB
18	51.967	52.443 dB	0.476 dB	± 0.25 dB